Math 344, S19 **Name:** Answer

Quiz 5

<u>Instructions:</u> There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Which of the following subsets of $\mathbf{R}^3$ are actually subspaces? (no explanation needed)

(a) The plane of vectors (b_1, b_2, b_3) with $b_1 = b_2$.

Yes

(b) The plane of vectors (b_1, b_2, b_3) with $b_1 = 1$.

No

(c) All vectors that satisfy $b_1 + b_2 + b_3 = 0$.

Yes

(d) All vectors that satisfy $b_1 \leq b_2 \leq b_3$.

No

(e) All vectors (b_1, b_2, b_3) that are not in the column space $C(\mathbf{A})$ where $\mathbf{A}$ is an 3×3 matrix.

No

Problem 2. (5 points) Find the column space $C(\mathbf{A})$ and the null space $N(\mathbf{A})$, where $\mathbf{A} = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix}$.

$$A = \begin{bmatrix} \boxed{1} & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ -2 & -6 & 0 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 2 & 4 \end{bmatrix} \to \begin{bmatrix} \boxed{1} & 3 & 1 & 2 \\ 0 & 0 & \boxed{2} & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = U$$

pivot columns/variables : x_1, x_3

free columns/variables : x_2, x_4.

$$C(A) = \left\{ c_1 \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 4 \\ 0 \end{bmatrix} \,\middle|\, c_1, c_2 \text{ any real numbers} \right\}.$$

For special solutions to $AX=0 \Leftrightarrow UX=0$

let $x_2 = 1$, $x_4 = 0$. we get

$$\begin{cases} x_1 + 3 + x_3 = 0 \\ 2x_3 = 0 \end{cases} \Rightarrow \begin{matrix} x_1 = -3 \\ x_3 = 0 \end{matrix} \Rightarrow s_1 = \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

let $x_2 = 0$, $x_4 = 1$. we get

$$\begin{cases} x_1 + x_3 + 2 = 0 \\ 2x_3 + 4 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = 0 \\ x_3 = -2 \end{cases} \Rightarrow s_2 = \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

Thus

$$N(A) = \left\{ c_1 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \,\middle|\, c_1, c_2 \text{ any real numbers} \right\}.$$